

How minor miner works:

It's a lot of dijkstra in a loop.

First it randomly places the first variable. Then for each additional variable it seeds its chain at an optimally costed location. This cost is the sum of shortest paths to the neighbor's chains, which is where dijkstra comes in. each chain seeding is a single-source shortest path for the entire graph for each of the variable's neighbor chains as sources. After the 'seed' is placed at the lowest cost qubit, the chain is formed from the union of shortest paths to each of the neighboring chains **<This is where claude suggests approximating the steiner tree problem to get better chains>**

Minor miner allows chains to overlap initially. To support this, the dijkstra costs aren't just raw distance, they're $\langle \text{graph_dimater} \rangle ^ \langle \text{chains_using_qubit} \rangle$ as its cost so that a long path of unused qubits is preferable to a short path where even one is taken, similarly a path of all used qubits is preferable to one where a single qubit is already claimed by two other chains. **<claude disputes this cost, thinks it could be replaced by 'negotiated congestion' from FPGA switch design>** the difference is that it keeps a history term of how often a qubit has been congested that increases with each iteration, eventually somebody must yield if they can. It might be problematic though, since if this mechanism resolves multiple conflicts, it might have to crank the second one up to a BIG number to get a reroute through an already inflated number, and then resolve that again by making it even BIGGER. Some sort of decay should probably happen once a qubit is decongested.

Lastly comes the order or resolution. Minor miner picks a fixed random vertex order, and removes and rebuilds the chains one at a time until the solution is feasible or it gives up. **<claude likes something called cuthill-mckee>** it wants to reduce something called 'bandwidth' in a graph. This quantity is the minimum length of an edge if the vertices of the graph were embedded uniquely at integers on a number line. It seems the hope is that by ordering to reduce bandwidth it tends to rebuild related chains in close proximity to each other, which would hopefully tend to help with congestion.